Vertically Integrated Industries

Initial takeaways on minerals, metals, manufacturing and chemistry

Smith Point Capital theme memo, v2, October 28, 2026. Drawn from our primer (v10.1), its sources and new research on Chemify and Warp Sciences. Company statements are labeled as claims.

What this theme is

This theme covers companies that make physical things in their own plants and use their own software and AI to make them faster, and with fewer skilled people, than older competitors. Their revenue comes from the part, metal or molecule they ship. We call them vertically integrated because one company owns the whole job, from taking the order to shipping the product.

Hadrian is the clearest example. It machines metal parts for rockets, missiles, satellites, jets and submarines in automated factories in California and Arizona. Its in-house software handles quoting, programming the machines, scheduling and quality checks. A customer sends a drawing and gets parts back. Lockheed liked the result enough to put Hadrian's cells inside its own missile plant.

Picture the supply chain as a ladder. Companies mine ore, then refine it into pure metals and products like magnets. Distributors cut metal to size. Fabrication shops cut, bend and weld it into frames and enclosures. Precision shops make tight-tolerance parts by computer-controlled machining (CNC), metal 3D printing or casting, mostly for aerospace and defense. Chemistry sits alongside: making the molecules that drug and materials programs need. At the top sit the primes, the few big contractors (Lockheed, RTX, Northrop, General Dynamics, Boeing) that sell finished weapons and aircraft to the government.

Why now? China has shown it will cut off critical minerals and magnets. The Pentagon needs far more missiles and ships than its suppliers can make. Washington is lending, buying and taking equity stakes to get US production built. And the owners of America's small machine shops are retiring with no one to replace them.

1. Key takeaways

1. The tailwinds are large and mostly lasting, but much of the money is conditional. Supply-chain security and the labor gap will outlast any single budget; tariffs and the munitions surge could fade.

Tailwind	Key numbers	Lasting or cyclical	Who it helps most
China's hold on refining and magnets	Leads refining in 19 of 20 strategic minerals at about 70% share (IEA); about 94% of rare-earth magnets (Mining.com); US fully import-reliant for 16 minerals (USGS)	Lasting. China's current pause on export controls ends Nov 10 and Nov 27, 2026	Magnet and metal makers; parts makers that need secure inputs
Munitions and shipbuilding	$49.5B for 10 munitions in the FY27 request, 8.5x the past decade's average, but 82% depends on an unpassed reconciliation bill (CNAS); Navy put $900M beside Hadrian's own money for submarine-part plants (Breaking Defense)	Need is lasting; funding timing is cyclical	Precision parts and structures (Hadrian, Divergent, Machina, Senra)

Tailwind	Key numbers	Lasting or cyclical	Who it helps most
Multi-year buying in law	FY26 defense bill allows multi-year munitions buys and purchase commitments of up to 10 years with advance payments (FY26 NDAA)	Lasting law; each year still needs appropriations	Producers that already have a plant running
Government lending and equity	Pentagon OSC loans of $620M to Vulcan and $500M (conditional) to Phoenix, about $20B requested for FY27 (Mintz); DOE loan office has over $289B of authority (DOE); $820.5M in 23 Defense Production Act awards since Jan 2025; $65B of public minerals money worldwide in 2025 (IEA)	Mixed. Large, but conditional and political	Capital-heavy producers that can carry debt
Labor gap	3.8M manufacturing jobs to fill by 2033, 1.9M may go unfilled (Deloitte); 83% of 17,156 US machine shops have under 20 staff (Census)	Lasting	Automated producers and buyers of old shops (Amca)
Drug-chemistry onshoring	BIOSECURE became law in Dec 2025; 79% of 124 biotechs surveyed had a China contract manufacturer (BIO); replacing those partners could take up to eight years (industry estimate)	Lasting policy, slow enforcement	US chemistry labs (Warp, Chemify)
Tariffs	50%, 25% and 10% tiers on metal goods since April 2026 (White & Case); Nucor's margin rose from 14% to 20%	Cyclical. Set by proclamation	Domestic mills first, then US parts makers
Reshoring and data centers	Fabricated-metal shipments rose from $471.5B in 2023 to $490.4B in 2025 (Census); about 247,000 reshoring jobs announced in 2025, but only 14 of 64 tariff-linked announcements had turned into projects by mid-2026 (Reshoring Initiative)	Partly cyclical. Pledges run ahead of orders	Fabricators (SendCutSend, 1872)

Three things push the other way. The Fed raised rates to 3.75-4.00% in September, and manufacturing construction spending was down 21.9% year on year by May 2026 (Federal Reserve, ENR). Tariffs raise input costs for the same parts makers they protect, and Nucor's own fabrication unit reported margin compression. And permanent bonus depreciation lets a profitable incumbent write off a $2M machine in year one, which makes automation cheaper for the competition we face.

The security and labor rows explain why buyers will keep paying for US capacity for a decade. The budget and tariff rows explain why 2026-27 feels hot, and they could cool. We want companies that would keep their customers if tariffs went away or China cut a deal.

2. A factory has to spend its capital before it wins the contracts. That is the central risk in this theme and the main thing we underwrite.

A prime or a drug company won't qualify you or sign a volume contract until your plant exists, your machines are running and your certifications are done. So the startup pays for all of that first, often two years before real revenue. Hadrian's CEO puts it bluntly: "no one's going to give you a production contract unless you've already started executing," and capex runs about two years ahead of revenue (claim, Relentless podcast). He adds that only customer prepayments or long-term infrastructure credit have worked to fund it.

Hadrian committed about $200M to a 270,000-square-foot plant in Mesa, Arizona (Plant Services). Its submarine-parts plants pair $900M of Navy money with more than $1.5B of Hadrian's own (Breaking Defense).

Divergent is building a 430,000-square-foot second factory in Long Beach with 64 of its own printers. About 80% of it is sized for defense volume, per the CEO, while Raytheon's Tomahawk contract still waits on the FY27 budget (company).

Vulcan Elements is building a 10,000-ton magnet plant in North Carolina with no commercial buyer named yet. Phoenix Tailings announced a $375M separation plant in Rochester, New York, in October 2026 with one $100M automaker contract and no public long-term mine supply. In chemistry, Chemify is building a second automated lab in Glasgow as part of an £89.9M expansion (The Scotsman), and Warp Sciences raised $215M to build US labs while working with only "a small number of design partners" (Warp).

This matters for venture returns because equity is the most expensive way to buy a building. Venture money spent on a plant buys an asset with asset-like returns, and founders give up a lot of ownership before the first volume order. If qualification slips a year, the plant sits half empty. A process plant needs about 80-85% utilization to earn its cost of capital (FTI Consulting). A hardware company at Series B often has only about $10M of revenue and break-even gross margins (Lowercarbon).

The best companies push the capex risk onto someone else:

- Government loans and grants. Vulcan's $620M OSC loan, Phoenix's conditional $500M OSC loan, the Navy's $900M with Hadrian and Chemify's £22M of UK grants all pay for buildings that equity would otherwise fund.
- Offtakes signed before the build. An offtake is a contract to buy a set amount of future output. MP Materials, the largest US rare-earth producer, got a Pentagon commitment to buy all output from its next magnet plant for 10 years, plus a price floor, before building it (MP Materials).
- Customer-hosted capacity. Lockheed hosts Hadrian's cells inside its own plant on a 7-year deal, so Hadrian skips the building. The FY26 defense bill now allows advance payments on purchase commitments.
- Bank debt once revenue exists. Hadrian added a $360M credit line in August 2026 (ABL Advisor).
- Leasing. Senra opened its second plant, about 80,000 square feet in Cypress, California, within months of signing the lease (company). The Pentagon did this in the 1950s, paying for 10 giant forging presses and leasing them to industry.
- Franchising. Isembard's franchisees buy their own machines and run on Isembard's software.
- Buying plants that are already certified. Amca buys qualified aerospace component makers, so the capex and the approvals already exist. One roll-up founder said they buy shops at about seven times EBITDA (Re:Build).

3. Buyers pay for speed and spare capacity. Few will pay for a lower price, and the speed premium covers only part of their work.

This was the clearest message from about 30 expert calls with buyers. A Northrop director said "Cost is secondary. Cost is driven usually by schedule." A Northrop engineer said he would pay 20-40% more to avoid a delay. Even Hadrian's CEO concedes that "most of the market just doesn't value speed."

The premium is real on prototypes, spares and urgent low-volume orders, and fades at volume. Buyers also cap each supplier's share. SpaceX keeps a shop at about a third of its revenue, and Hadrian holds 2-3% of one Boeing unit's precision spend today (expert calls).

Cheap labor matters less than people think. In a new automated plant it is under 10% of the cost of goods (claim, Drew Baglino on a16z). China wins because every supplier is a short drive away. The pitch that sells is "we deliver when your current shop can't," and it works best in defense, where rules already keep work in the US.

4. The margin goes to whoever owns the output and is approved to supply it.

Owned factories earn more than networks of partner shops. Proto Labs makes 49% gross margin in its own plants and 31% on work it farms out (Protolabs). Howmet's engine unit runs a 37.7% EBITDA margin and Carpenter 32.1% (company filings). A software vendor selling to these plants never touches that margin. Approval is the other half. Once a supplier is qualified on a program that can run 50 years, it tends to stay (Boeing expert call). That is why buying a certified plant, as Amca does, can be worth more than building a new one.

5. AI's real edge today is in the office work around the machine and in processes that run nonstop. Nobody has yet shown software making each unit cheaper as volume grows.

AI already speeds up quoting, machine programming, work instructions and quality checks. Arda's software cut assembly-instruction writing at Anduril from about 10 hours to 15 minutes (claim). A former Hadrian programming lead said time from order to first part fell about 50% over two years. That is the only learning-over-time number in our sources.

The data advantage is strongest where a process throws off constant measurements, like a metal printer watching every layer (Freeform), a refinery retuned every 15 minutes (Mariana, claim), or a robotic chemistry lab that records every reaction, including the failures most labs never write down (Warp, Chemify). Batch machining has less of it. A Boeing strategy head put the net saving from automation, after paying for the machines, at 20-30% at best.

There is also a software model that fits us. In a sold-out process plant, each extra point of yield is close to pure profit, and European plants average about 74% utilization (FTI Consulting). Software that runs the plant and is paid on the extra output earns like a producer and skips the capex problem in takeaway 2.

6. The biggest risk for a new producer is that approved incumbents automate first and big buyers keep the work in-house or spread it thin.

Three groups hold the power: the primes, which decide who gets qualified; the mining majors (BHP, Rio Tinto) plus MP in rare earths; and approved suppliers like Howmet and Precision Castparts (PCC).

The incumbents are moving. PCC's share of Boeing machining fell from about 50% to 30% over ten years, and it is now automating (expert call). The majors mostly buy innovation. BHP Ventures backs Mariana, Rio paid $6.7B for Arcadium Lithium, and RTX Ventures led Radical AI's round. In chemistry, big pharma can build its own robotic labs. Eli Lilly and NVIDIA plan to put up to $1B into a joint AI lab (Lightspeed).

Failures show what happens when the speed story outruns the economics. Daedalus, an AI-run CNC factory in Germany, went insolvent in August 2026, and Markforged, worth $2.1B in 2021, sold for $42.5M (3D Printing Journal). The survivors will sell to several primes and to commercial buyers too.

2. Where we would invest

Precision parts now come first. The leaders there have the most proof of qualified, paid production, and at $1B to $8B they are within the range we will consider. Fabrication is next, then plant software, which fits an enterprise software fund best. Chemistry labs and metals follow because they lean on policy and carry the most capex risk.

A. Precision and defense parts

How it ties to takeaways 2, 3 and 4: these companies sell speed into programs that last decades, and each has a different answer to the capex problem. Hadrian borrows and co-invests with the Navy, Amca buys plants that already exist, Divergent builds its own printers, Isembard franchises and Freeform sells printed parts from a single plant.

Incumbents leave room because the lower tiers are huge and untracked. The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of F-35 sub-tier suppliers (GAO).

Hadrian ($1.37B Series D, August 2026, at $7.87B; anchored by JPMorgan's Strategic Investment Group; about 643 staff)

- Overview and use cases: Hadrian makes CNC-machined metal parts for rockets, missiles, satellites, jets and, from 2026, Virginia- and Columbia-class submarines. Its factories in Torrance, California and Mesa, Arizona run on its own Opus software, and it also runs cells inside customer plants such as Lockheed's. Customers include Lockheed, RTX, Anduril and the Navy.
- Thesis: Hadrian has more proof than anyone in the theme that primes will qualify and pay a new producer, and it has shown the capex playbook in takeaway 2: Navy co-investment, a Lockheed-hosted cell deal and a $360M credit line. It is the strongest business in our set, and our help would be introductions to commercial buyers outside defense.
- Areas to diligence: The price is $7.87B, so we need a view on margins. A former employee called it "not … a fantastically profitable business." We would test whether Opus lowers unit cost over time and how Hadrian grows past the 25-30% share buyers allow. Its CEO says some Chinese parts land at about a third of his cost (claim). Access to an allocation is also to confirm.

Amca ($300M Series B at over $1B, led by Caffeinated Capital and Lightspeed; revenue to confirm)

- Overview and use cases: Amca buys certified aerospace and defense component makers and modernizes them. It owns eight plants, including Electrocube in Pomona, California, that make hydraulic and electrical parts, switches, valves, transformers and power supplies for the F-35, F-16, F/A-18, Mk-48 torpedo and M1 Abrams. Its AI platform, RAPID, handles engineering and requalification across the plants. It claims an Army Materiel Command contract and work with over a dozen primes.

- Thesis: Amca answers takeaway 2 directly. It buys capex and qualification that already exist, often for single-source parts the primes can't easily move, and adds software to raise margins (takeaway 4). A buy-and-improve platform suits our operating and partnership support.
- Areas to diligence: Revenue, EBITDA and the multiple paid per plant are to confirm. We would test whether RAPID actually lifts margins or whether returns come from buying cheaply, how much debt sits on the plants, and whether key engineers stay after each deal.

Divergent ($290M Series E, September 2025, at $2.3B; over $1B raised; about 500 staff)

- Overview and use cases: Divergent designs and 3D-prints metal structures for missiles, aircraft and cars, then assembles them with robots and no fixed tooling. It builds its own printers, which it does not sell, and charges per part. Named customers include Lockheed, RTX, General Atomics, Saab, Bugatti and McLaren. It is a second source for Raytheon's Tomahawk midbody, with production aimed for early 2027.
- Thesis: At $2.3B Divergent is well inside our range. It owns everything from design software to printers to factory, sells into both defense and cars, and claims 2025 revenue grew more than 5x (claim). That two-market mix is a hedge against the budget risk in takeaway 1.
- Areas to diligence: The Long Beach factory is the biggest capex-before-contract bet in our set, since it is mostly sized for missile volume that waits on the FY27 budget. Cost parity with casting is only the CEO's claim. We would also confirm whether there has been a newer priced round, because $2.3B is from September 2025.

Isembard ($50M Series A, March 2026, led by USV; claims 14 factories)

- Overview and use cases: Isembard machines precision parts from aluminum, steel, titanium and Inconel (a heat-resistant nickel alloy) for jet engines, drones, satellite antennas and robots. Named customers include Anduril, Babcock, Tekever and Oxford Space Systems. It runs owned and franchised factories on one system, MasonOS, from a London headquarters with sites in the US, Germany, France and Ukraine. It claims 14 factories (4 owned, 10 franchised) and 10x revenue growth year on year.
- Thesis: Franchisees pay for the machines and Isembard supplies software, customers and standards. That is the cleanest light-capex answer to takeaway 2. It claims it cut one customer's lead time from about 10 weeks to under 14 days at the same price. In September 2026 it said it expects to raise significant additional funding, so it is actionable now.
- Areas to diligence: We would ask for economics per franchise: what a franchisee earns, what Isembard keeps and how fast a site turns a profit. A Thales Alenia Space procurement head doubted primes will accept quality across franchised sites.

Freeform ($67M Series B, February 2026; about $179M post-money per PitchBook)

- Overview and use cases: Freeform 3D-prints metal parts that carry fluid and heat, like heat exchangers, manifolds and rocket-engine injectors, for space, defense, energy and chip-equipment customers (claim). Its own multi-laser printers use cameras and AI to correct each

layer as it prints. It sells the parts and keeps the printers. It reports hundreds of parts a week and has a Defense Innovation Unit contract.

- Thesis: Freeform has the best data story in manufacturing, because its process logs sensor data on every layer (takeaway 5). It is the cheapest entry in this area.
- Areas to diligence: Customers are undisclosed, so we need reference calls. Yield trends over time would prove the data story and we haven't seen them. Whether its large new printer is running is unconfirmed.

B. Fabrication and assembly

How it ties to takeaways 2 and 3: these companies sell speed on high-mix work (many small, different orders) and their plants cost tens of millions. Incumbents leave room because the shops doing this work are small, aging and short of programmers. Their machines cut metal only about 34% of the time against a 70% target (industry survey of 100+ plants).

Senra Systems ($65M Series B, July 2026, co-led by Lowercarbon and Interlagos; about 190 staff)

- Overview and use cases: Senra makes wire harnesses and cable assemblies, the bundles of wires and connectors that carry power and signals through aircraft, satellites and defense systems. It builds them in two plants in Redondo Beach and Cypress, California. Its software, Amp, turns a customer's design into a digital model and step-by-step instructions for technicians. It claims to have cut harness lead time from six months to two weeks, and says 99% of harnesses pass inspection the first time.
- Thesis: Harnesses are still made by hand, and a late one holds up the whole aircraft or missile, so customers pay for speed (takeaway 3). Senra leased its second plant and opened it within months, which is the light-capex approach we want (takeaway 2). It just raised, so this is a relationship to build for the next round.
- Areas to diligence: We would check the yield and lead-time claims with customers, gross margin by program and how concentrated repeat revenue is.

SendCutSend (Series B at $2.1B, October 2026, after a $1.01B round in May 2026)

- Overview and use cases: SendCutSend cuts, bends and finishes custom sheet-metal parts that customers order online, in plants it owns. It handles about 32,000 orders a month and has been profitable for eight years.
- Thesis: This is the proof that an owner-operator in fabrication can be profitable at scale, and it was bootstrapped, so the capex came from earnings (takeaway 2). It is under our $10B ceiling.
- Areas to diligence: The price doubled in five months, and the $2.1B figure comes from a single trade report, so it is to confirm. Customers tend to "outgrow" online fabricators, and the founder says he beats China on only about 20-25% of his work (claim).

Machina Labs ($124M Series C, February 2026)

- Overview and use cases: Machina shapes sheet metal into missile airframes and aircraft and car panels with pairs of robot arms, skipping the costly dies a stamping press needs. It claims its first plant will make thousands of missile airframes a year.
- Thesis: A new die takes months and a lot of money, so skipping it suits defense programs whose designs change often (takeaway 3). The founder says Machina won't start a project unless the customer commits dollars, a good capex discipline.
- Areas to diligence: Each part takes 30 minutes to hours, against seconds on a press (MMS). We would test where it wins at volume, verify the claimed 50x lead-time gain, and split paid production from development work.

We are also tracking 1872 ($15M seed via Ohio's O.H.I.O. Fund), which uses robots to weld steel skids and enclosures for grid and data-center customers in Cincinnati. It has no customer evidence yet.

C. Software that runs plants

How it ties to takeaways 2 and 5: this is the way to back the theme without carrying a plant. Most enablers are at first-customer stage, but one model fits us: software that runs the plant and is paid on what it adds. Incumbents leave room because Rockwell, Siemens and the process-control vendors (Honeywell, AspenTech, Emerson) sell hardware first or advise operators.

Lumaril ($12M seed led by Lux; a WSJ Pro headline reports a $66M round at $400M)

- Overview and use cases: Lumaril connects to a process plant's existing controls, adds cheap audio and video sensors, builds a live model of the plant and adjusts setpoints itself, through a safety layer, to raise throughput and yield. A setpoint is a target such as a temperature or flow rate. It says "We're paid on the production we add" (claim). The founders built Palantir Foundry, Palantir's commercial data platform.
- Thesis: Lumaril earns like a producer with no capex, and because it runs the plant directly it builds the data a plant owner would (takeaway 5). The buyers are large industrial enterprises, and the founders' background fits a big-account sales motion where our partnership help counts.
- Areas to diligence: Customers and results aren't public, so we need one plant's before-and-after yield. We would ask how the outcome-pricing baseline is set and how a plant's safety team approved software that adjusts the plant on its own. The round size comes only from a paywalled headline.

Arda ($160M Series A at $1.5B) is the factory-side comparable. Its AI turns 3D designs into work instructions, but Anduril is its only named user.

D. AI-run chemistry labs

How it ties to takeaways 1, 2 and 5: these are the only two chemistry companies we want in the theme, and neither makes bulk or specialty chemicals. Both build robotic labs that make small batches of new molecules for drug and materials programs, then learn from every run. Drug companies send most of

this work offshore, largely to China, and the two biggest offshore providers are worth close to $100B combined (Lightspeed). BIOSECURE and the Pentagon's June 2026 listing of the largest provider as a Chinese military company push buyers toward a US option. Both companies also carry the capex-first risk in takeaway 2.

Chemify (Series B of £51.9M, about $70M, co-led by Wing and Insight Partners; about £110M raised including grants; valuation to confirm)

- Overview and use cases: Chemify, a University of Glasgow spin-out founded by chemist Lee Cronin, turns molecule designs into code that robots run to make the compound. It calls this "chemputation," built on its own chemistry programming language and a large library of tested reactions (Insight Partners). Its first automated lab, the Chemifarm, opened in Glasgow in June 2025 and makes molecules at milligram-to-gram scale (claim, SCI). It mainly sells discovery chemistry to pharma: designing, making and refining drug candidates. Named partners include Sanacor (heart disease) and the Gates Foundation (tuberculosis). Cronin claims it works with eight of the top 20 pharma companies (SCI). Staff are about 152.
- Thesis: Chemify owns the lab, the language and the reaction data, so its customers' experiments improve its own system (takeaway 5). It pushed part of its capex onto government, with £22M of Scottish and UK grants toward the second Chemifarm (The Scotsman). It plans a US Chemifarm, where our go-to-market and pharma introductions could help.
- Areas to diligence: Revenue is undisclosed, and the "eight of the top 20" claim may include pilots. Milligram-to-gram output means this is research services for now. We would ask whether the US lab is funded and how it competes with pharma's in-house robotic labs.

Warp Sciences ($215M Series A, October 2026, led by Lightspeed; $865M post-money per its CEO)

- Overview and use cases: Warp is a San Francisco-based chemistry contract research organization (CRO), meaning a lab that does chemistry work for drug companies under contract. Its first service is lead optimization: making the many variations of a promising drug molecule that a customer's scientists compare before choosing a candidate. Expert chemists do most of the work at first, with AI choosing the next experiment and robots taking over repetitive steps over time. It records every step, including failed reactions, to train its models, and plans to sell predictions as well as compounds later (Lightspeed). It does not develop drugs of its own. Founders include Kevin Weil, who led science at OpenAI, and DJ Patil, the former US chief data scientist.
- Thesis: Warp targets the exact work BIOSECURE pushes back to the US (takeaway 1), and its business model sells the output, compounds, while keeping the data (takeaways 4 and 5). Its pharma buyers are enterprise accounts where our partnership work applies.
- Areas to diligence: There are no named customers yet, only design partners, and it is valued at $865M at launch. It is building labs ahead of contracts (takeaway 2). Matching offshore prices with US labor is hard, and lead-optimization batches are tiny, so we would test revenue per customer. Pharma's own AI labs are the main competitor.

E. Rare-earth metals and magnets (selective, at most one investment)

How it ties to takeaways 1 and 2: China's hold on magnets is the most lasting limit in the theme, but these plants carry the most capex before contracts. We would only back a company where government debt pays for the plant, a buyer already pays, and our money goes to software and ramp-up. Incumbents leave room because MP is the only scaled US producer and still lost $85.9M on $224.4M of revenue in 2025 (MP Materials).

Vulcan Elements ($65M Series A, led by Altimeter; $620M OSC loan)

- Overview and use cases: Vulcan makes NdFeB magnets (named for neodymium, iron and boron) for drones, chip tools, robots and cars from non-Chinese material. It buys refined oxides from ReElement, has a supply deal with Ucore from 2027 and is building a 10,000-ton plant in Benson, North Carolina. The Army chose it for its SkyFoundry drone program.
- Thesis: A defense buyer already orders its product and government debt pays for the plant, which is the right capex structure (takeaway 2). It claims its pilot output is sold out a year ahead.
- Areas to diligence: Federal awards total only $5.7M across seven contracts (USAspending). We would test unit cost against Chinese magnets and whether commercial buyers sign without subsidy.

Phoenix Tailings ($116.6M Series B in total; $500M conditional OSC loan)

- Overview and use cases: Phoenix turns rare-earth concentrate and recycled magnet scrap into the metals and alloys magnet makers use. It runs a 200-ton-a-year plant in Exeter, New Hampshire, and is building the Rochester separation plant to make over 4,400 tons of oxides a year from 2027. It has a $100M contract with an undisclosed automaker.
- Thesis: Phoenix sits at the step where China is strongest, between ore and magnet, and is the most advanced US rare-earth metal maker after MP (takeaway 1).
- Areas to diligence: It plans to grow about 15 times in two years. The OSC loan is conditional, a $66M DOE grant is only a selection, and no long-term mine supply or price floor is public.

Mariana Minerals ($310M Series B, led by Khosla; valuation to confirm)

- Overview and use cases: Mariana restarted a copper mine in Utah in May 2026 and refines it into 99.99% pure copper (claim), and is building a plant to pull lithium from Texas oilfield wastewater.
- Thesis: It has the strongest data story in metals and owns its output end to end (takeaway 5).
- Areas to diligence: No copper buyer is named, the lithium buyer is unnamed (claim), and the check size is likely above ours.

F. Watch only: materials discovery labs

Radical AI ($55M led by RTX Ventures), CuspAI ($450M Series B) and Periodic Labs ($300M seed) use AI and robot-run experiments to invent materials. None sells material yet, and a new material still faces

one to five years of buyer qualification. Radical, with a defense backer, is likeliest to sign a supply deal first.

Recommendation

Our most actionable names are Hadrian, Isembard and Lumaril, with Amca and Senra close behind.

Hadrian has already solved the capex problem with Navy money, a Lockheed-hosted deal and bank debt. The question is price and access at $7.87B. Isembard has the lightest capex model in manufacturing and says it expects to raise soon. Lumaril fits an enterprise software fund best, paid on output with no plant.

Amca is the cleanest way to buy existing approvals, and Senra the cleanest owner-operator at an early price. In chemistry, Chemify is further along and Warp is better funded. We would meet both before forming a view.

3. Where we would not invest

We would avoid layers where returns depend more on financing, commodity prices or government programs than on the help we bring.

- Exploration and new mines (Earth AI is the closest name). US mines take 15.7 to 19 years from discovery to production, and Earth AI has yet to publish a formal resource estimate. We would watch it.
- Greenfield refineries and big metal plants funded mostly with equity. Each is one giant bet better suited to government debt.
- Metal distribution. Incumbents like Reliance are gaining share without AI, and the value sits in inventory, credit and relationships.
- Bulk and specialty chemical producers. They fight on commodity prices and long regulatory reviews. Our chemistry exposure comes through Chemify and Warp.
- Companies valued above about $10B, where we can’t own enough to matter.
- Discovery labs with no product (Periodic, Lila Sciences). Valued in the billions before selling anything.
- General factory robotics (Mind Robotics). A $3.4B valuation with one main customer, Rivian, which owns 32.1% of it.

4. What we would underwrite

We would back companies that own their output, get paid for speed, and have moved the cost of their plants onto someone else. In practice that means five tests.

1. Committed demand before the build: a signed offtake, a prepayment, a customer-hosted deal or a qualified buyer already ordering.
2. Debt, government loans or grants, customer money, leases, franchisees or an acquired plant pay for most of the capex, so our equity funds software, people and ramp-up.

3. It owns production or the outcome of production, through its own floor, a franchise network on its software, or pricing tied to the output it adds.
4. It sells to more than one big buyer and holds approvals that take years to repeat.
5. It can show with numbers that its software improves yield, lead time, scrap or cost over time.